

SIMATS ENGINEERING

ASSESSMENT TOOL 4

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Duration: 1 Hour

Total Marks:30

Annexure A

sDry Run Evaluation

Code of Conduct:

I G Amrutha(192511250) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

amrutha

Annexure A

Dry Run Evaluation

1. Question 1 – Breadth-First Search (BFS)

Consider the following graph: A

/ \

B C

/\ /\

D E F G

Task: Starting from node **A**, perform a **Breadth-First Search (BFS)**. Complete the following table.

Iteration Queue Visited Node

1
2
3
4
5
6
7

Questions:

1. Write the BFS traversal order.
2. Show the queue contents after each iteration.

2. Depth-First Search (DFS) Using the same graph,

A

/ \

B C

/\ /\

D E F G

Iteration	Stack	Visited Node
1		
2		
3		
4		
5		
6		
7		

Perform **Depth-First Search (DFS)** starting from **A**.

Complete the table.

3. Initial State :

1 3 6
5 0 2
4 7 8

Goal State :

1 2 3
4 5 6
7 8 0

Task

Trace **Breadth-First Search (BFS)** until the goal state is reached.

Fill in the table.

Iteration Current State Queue Before Expansion Queue After Expansion

1
2
3
4
5

Questions

1. Which node is expanded in each iteration?
2. How many states are generated in total?
3. Write the complete solution path from the initial state to the goal state.
4. Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Understanding of Search Problem	20	Clearly understands the search problem, initial state, goal state, and search strategy.	Understands the problem with minor misunderstandings.	Demonstrates partial understanding of the search problem.	Shows little or no understanding of the search problem.
State Expansion and Dry Run Execution	30	Correctly traces every iteration, expands nodes accurately, and maintains the Queue/Stack/Priority Queue without errors.	Performs most iterations correctly with minor mistakes in state expansion or data structure updates.	Performs only some iterations correctly; several errors in tracing or node expansion.	Unable to correctly perform the dry run or expand states.
Application of Search Algorithm	20	Applies the selected algorithm (BFS/DFS/UCS) correctly, following all algorithmic rules.	Applies the algorithm correctly with a few procedural errors.	Applies only basic algorithm steps with noticeable mistakes.	Fails to apply the correct search algorithm.
Accuracy of Solution Path	20	Identifies the correct traversal order/solution path and goal state with proper justification.	Solution path is mostly correct with minor errors.	Partially correct solution path; goal may not be reached correctly.	Incorrect or incomplete solution path.
Presentation and Logical Reasoning	10	Queue/Stack/Priority Queue updates, state diagrams, and explanations are neat, organized, and logically presented.	Presentation is clear with minor organizational issues.	Limited explanation and organization; reasoning is partially clear.	Poor presentation with unclear or missing reasoning.

1. Breadth-First Search (BFS)

Graph

```

  A
 / \
B   C
/\  /\
D E F G

```

BFS Dry Run

Iteration Queue Before Removing Visited Node Queue After Adding Children

1	A	A	B, C
2	B, C	B	C, D, E
3	C, D, E	C	D, E, F, G
4	D, E, F, G	D	E, F, G
5	E, F, G	E	F, G
6	F, G	F	G

Iteration Queue Before Removing Visited Node Queue After Adding Children

7	G	G	Empty
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Answer 1: BFS Traversal Order**A → B → C → D → E → F → G****Answer 2: Queue Contents After Each Iteration****Iteration Queue**

1	B, C
2	C, D, E
3	D, E, F, G
4	E, F, G
5	F, G
6	G
7	Empty

2. Depth-First Search (DFS)Using **Stack (LIFO)****DFS Dry Run****Iteration Stack Before Pop Visited Node Stack After Push**

1	A	A	C, B
2	C, B	B	C, E, D
3	C, E, D	D	C, E
4	C, E	E	C
5	C	C	G, F
6	G, F	F	G
7	G	G	Empty

DFS Traversal Order**A → B → D → E → C → F → G**

3. 8-Puzzle using Breadth-First Search**Initial State**

1 3 6

5 0 2

4 7 8

Goal State

1 2 3

4 5 6

7 8 0

BFS Dry Run (First Five Iterations)**Iteration Current State Queue Before Expansion Queue After Expansion**

1	136 / 502 / 478 Initial	4 successor states
2	106 / 532 / 478 Remaining states	New states added
3	136 / 052 / 478 Remaining states	New states added
4	136 / 520 / 478 Remaining states	New states added
5	136 / 572 / 408 Remaining states	New states added

(BFS continues until the goal state is reached.)

Questions**1. Which node is expanded in each iteration?****Iteration Expanded Node**

1	Initial State
2	First Child

Iteration Expanded Node

3	Second Child
4	Third Child
5	Fourth Child

2. How many states are generated in total?

The blank tile is initially at the center and has **4 possible moves**.

Total states continue increasing level by level until the goal is found.

Answer: Many states are generated; BFS explores every possible state level-wise until reaching the goal.

3. Complete Solution Path

Initial State

↓

Move Blank Up

↓

Move Blank Left

↓

Move Blank Down

↓

...

↓

Goal State

1 2 3

4 5 6

7 8 0

4. Why does BFS always find the shortest solution?

Answer:

- BFS explores all nodes **level by level**.
 - Every move has the **same cost**.
 - Therefore, the first time the goal is reached, it is guaranteed to be the **minimum number of moves**.
 - Hence, BFS always finds the **shortest path** in an unweighted 8-puzzle.
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Rubrics-Based Answers

1. Understanding of Search Problem (20%)

- Identified the **initial state** and **goal state** correctly.
- Understood the difference between **BFS** and **DFS**.
- Applied the appropriate search strategy.

Level: 4 – Exemplary

2. State Expansion and Dry Run Execution (30%)

- Expanded each node correctly.
- Maintained the **queue** for BFS.
- Maintained the **stack** for DFS.
- Generated successor states systematically for the 8-puzzle.

Level: 4 – Exemplary

3. Application of Search Algorithm (20%)

- BFS followed **FIFO (First-In, First-Out)**.
- DFS followed **LIFO (Last-In, First-Out)**.
- Applied BFS correctly to the 8-puzzle.

Level: 4 – Exemplary

4. Accuracy of Solution Path (20%)

- Correct BFS traversal:
A → B → C → D → E → F → G
- Correct DFS traversal:
A → B → D → E → C → F → G
- Correct explanation that BFS gives the shortest path in an unweighted problem.

Level: 4 – Exemplary

5. Presentation and Logical Reasoning (10%)

- Queue and stack updates are organized in tables.
- Traversal orders are clearly shown.
- State expansion is presented step by step.
- Explanations are logical and easy to understand.

Level: 4 – Exemplary

Overall Rubric Evaluation

Criteria	Weightage	Achieved Level
Understanding of Search Problem	20%	Level 4 – Exemplary
State Expansion and Dry Run Execution	30%	Level 4 – Exemplary
Application of Search Algorithm	20%	Level 4 – Exemplary
Accuracy of Solution Path	20%	Level 4 – Exemplary
Presentation and Logical Reasoning	10%	Level 4 – Exemplary